# Lesson 10: Build and present a small game

Pygame CE • S3 onwards • one or more sessions

**Learning intention**
Apply the course skills to plan and create a game for someone else to play.

## Choose a project

- **Collect:** gather objects before time runs out.
- **Dodge:** avoid moving obstacles for as long as possible.
- **Maze:** reach an exit without touching the walls.

## Project checkpoints

1. Sketch the screen and describe the player's goal.
2. Build a playable version with movement.
3. Add a challenge, such as targets, obstacles or a timer.
4. Add feedback, such as a score, sound or game-over screen.
5. Ask someone else to test it, and use their feedback to improve the game.

## Planning sheet

| | |
|---|---|
| **Game name** | |
| **Player goal** | |
| **Controls** | |
| **What makes it hard?** | |
| **How does the game end?** | |
| **Features I will add** | |

## Idea starter: moving obstacles

```
obstacles = [pygame.Rect(random.randint(0, WIDTH - 30), -30, 30, 30) for _ in range(5)]

# each frame:
for ob in obstacles:
    ob.y += 200 * dt
    if ob.top > HEIGHT:
        ob.topleft = (random.randint(0, WIDTH - 30), -30)
    if ob.colliderect(player_rect):
        state = "game_over"
    pygame.draw.rect(screen, (220, 80, 80), ob)
```

## Assessment criteria

| Area | What I am looking for |
|---|---|
| Functionality | The game starts, can be played, and ends or restarts correctly. |
| Programming | Movement, conditions, collision detection and variables are used appropriately. |
| Testing | Problems are found and fixed; feedback from a tester is used. |
| Communication | Can explain how part of the game works and suggest one improvement. |

## Support and stretch

- **Support:** offer a choice of project and a partly completed codebase.
- **Stretch:** add multiple levels, increasing difficulty, or a short design explanation.

## Present and reflect

- Demonstrate your game to a partner. What did they find hard or confusing?
- What is one thing you would improve next?
- Which Pygame skill are you most proud of?